

Ly thuyet trong tam

- AI, ML, Deep Learning, GenAI, LLM nam o au trong cung mot he?. AI la chiec o lon nhat bao gom moi he thong thong minh. ML la hoc tu du lieu, DL la mang no-ron nhieu tang, GenAI sinh noi dung moi, LLM la model nen chuyen ngon ngu. LLM khong phai toan bo AI nhung la tang nen cua hau het trai nghiem AI hien nay. [trang 3]
- Ba nhom AI chinh: phan loai sinh noi dung hanh ong. Discriminative AI gioi phan loai, du oan. Generative AI sinh ra thu moi. Agentic AI nhan muc tieu roi tu lam nhieu buoc. LLM la engine chung cua ca Generative lan Agentic. [trang 4]
- Lich su AI 70 nam: tu he chuyen gia en Transformer va ChatGPT. 1980: He chuyen gia \(\text{expert system}\) tap trung giai tot mot mien hep. 2009: Fei-Fei Li va ImageNet du lieu lon anh bai thuat toan khon. 2017: Transformer cho phep moi tu nhin sang cac tu quan trong khac. 2022: ChatGPT ua LLM en ai chung. [trang 5, 6, 7, 8]
- Ben trong LLM: Token, Context, Attention. Token la manh chu model oc, moi thu eu quy ra token va co gia. Context la ban lam viec co han, model chi nhin uoc luong chu nhat inh. Attention cho phep moi token chu ong nhin lai cac token khac, cham iem muc o lien quan. [trang 11, 12, 13, 14]

Ca lop thuong hoi

- Phan biet AI, ML, DL, GenAI, LLM. LLM la mot phan cua GenAI, GenAI la mot phan cua Deep Learning, Deep Learning la mot phan cua Machine Learning, Machine Learning la mot phan cua AI. LLM la model nen chuyen ngon ngu, khong phai toan bo AI. \(\text{1 hoc vien, trang 3}\)
- Co che Attention va quan ly Context. Attention co han va co iem mu. at thong tin quan trong o au hoac cuoi prompt, tranh e o giua. Context cang dai cang ton token va cham. \(\text{1 hoc vien, trang 14, 15, 16}\)
- Token va chi phi. Tokenizer cat van ban thanh manh nho. Tieng Viet co dau thanh, code co ky tu ac biet nen bi cat nho hon, dan en nhieu token hon. Moi token eu co gia. \(\text{1 hoc vien, trang 13}\)

Cau tukiem tra

1. LLM thuoc nhom AI nao?

Dap an: D. [trang 4]

2. Su kien nao anh dau buoc ngoat ua Transformer tro thanh nen tang cho LLM?

Dap an: C. [trang 8]

3. Tai sao model co the 'quen' thong tin o giua mot prompt rat dai?

Dap an: B. [trang 14, 16]